ASHWIN XAVIER FRANCIS

+1 (720) 879-9675 | Ashwin.Francis@colorado.edu | LinkedIn | GitHub

EDUCATION

University of Colorado Boulder *Aug 2025 – Expected May 2027*
Boulder, Colorado, USA
Master of Science in Aerospace Engineering Sciences, Astrodynamics and Satellite Navigation Systems

University of Petroleum and Energy Studies *Sep 2020 – May 2024*
Dehradun, India
Bachelor of Technology in Aerospace Engineering (Avionics)

RESEARCH AND ENGINEERING EXPERIENCE

Structures Advisor/ Subsystem Co-Lead & PEC Lead– Great Lunar Expedition for Everyone (GLEE) *May 2026 – Present*
Colorado Space Grant Consortium (COSGC), University of Colorado Boulder, USA

- Co-led Hardware Deployment Mechanism (HDM) development within a 17-member engineering team for a NASA-funded, student-led lunar mission aiming to deploy over 100+ LunaSats across the lunar surface, enabling spatially distributed measurements of lunar magnetic fields, surface vibrations, temperature, and regolith properties.
- Developed and bench-tested microcontroller-based C/C++ firmware to control lead-screw elevation, barrel rotation, and yaw positioning, integrating stepper-motor drivers and troubleshooting inconsistent actuator movement.
- Implemented dual-servo deployment controls with independent and coordinated operation, including arming, firing, and reset sequences.
- Supported hardware integration and functional testing of motor-driven deployment mechanisms, investigating actuator timing, electrical interfaces, and mechanical alignment.
- Contributed to subsystem design reviews and verification planning, while advising undergraduate team members on hardware development and troubleshooting.

Electrical Lead – Flying Lab for Airborne Isotope Research (FLAIR) *Aug 2025 – May 2026*
Institute of Arctic and Alpine Research (INSTAAR), University of Colorado Boulder, USA

- Served as Electrical Lead within a 10-member engineering team developing a UAV-mounted laser spectroscopy payload to measure oxygen-isotope($\delta^{18}O$) ratios in atmospheric water vapor, supporting investigations of moisture transport and atmospheric water-cycle processes.
- Designed and revised electrical schematics and PCB layouts for microcontroller control, power distribution, sensor interfaces, and communication between payload components.
- Redesigned power regulation using a buck converter and linear regulator, addressing thermal inefficiencies and improving electrical stability during integrated operation.
- Integrated bidirectional logic-level shifting for communication and interfaced the Vaisala sensor, laser electronics, and fan-control system.
- Supported electrical verification and system integration, including sensor characterization, power stability testing, and UAV avionics interface validation.

Instruments Lead – Heliospheric–Ionospheric Unified Twin Explorer for Responses (HUNTER) *Aug 2025 – Dec 2025*
University of Colorado Boulder, USA

- Served as Instruments Lead in a 5-member team developing a Pre-Phase A, twin-spacecraft heliophysics mission concept to investigate how solar-wind dynamic-pressure variations affect Earth's high-latitude ionosphere and thermosphere.
- Evaluated a scientific payload comprising fluxgate magnetometers (FGM), electric-field instruments (EFI), Langmuir probes (LP), and ion and neutral mass spectrometers (INMS) to characterize plasma, electromagnetic fields, and neutral populations.
- Developed instrument-level requirements through the Science Traceability Matrix, assessing measurement ranges, sensitivity, sampling cadence, technology readiness levels (TRLs), and size, weight, and power (SWaP) constraints.
- Investigated payload accommodation and integration considerations, including sensor placement, magnetic cleanliness, calibration, and science data-handling requirements.
- Conducted green monopropellant trade studies for orbital phasing and formation maintenance; contributed to C&DH design, SpaceWire interfaces, GNSS synchronization, onboard storage, and S-band communications.

PROJECTS

Minimum-ΔV Impulsive Station-Keeping Near Sun–Earth L_4 and L_5 *Aug2026 – Present*
University of Colorado Boulder, USA

- Formulating a fuel-optimal station-keeping strategy to determine the magnitude, direction, and allocation of impulsive velocity corrections near the Sun–Earth L_4 and L_5 equilibrium points.
- Optimizing corrective maneuver magnitudes and directions to minimize cumulative ΔV while satisfying prescribed spacecraft position constraints over a fixed time horizon.
- Examining the dynamical symmetry through comparative analysis of fuel-optimal maneuver requirements and trajectory responses under appropriately symmetric initial perturbations.

Statistical Orbit Determination Using Ground-Based Tracking *Aug2026 – Present*

University of Colorado Boulder, USA

- Developing a MATLAB-based nonlinear orbital dynamics and state transition matrix (STM) propagation framework for spacecraft state estimation in low Earth orbit and modeling a network of 12 rotating ground stations, incorporating visibility constraints and noisy range, range-rate, and bearing measurements.
- Implementing Linearized and Extended Kalman Filters (LKF/EKF) to estimate spacecraft position and velocity, evaluating estimation errors, uncertainty propagation, and filter convergence.

Spatial and Temporal Sensitivity in Particle Motion near Lagrange Points *Jan 2026 – May 2026*

University of Colorado Boulder, USA

- Investigated spacecraft motion near the Sun–Earth L_4 and L_5 equilibrium points using the Circular Restricted Three-Body Problem (CR3BP), numerically propagating nonlinear trajectories to examine sensitivity to initial-state perturbations.
- Applied state transition matrices to analyze perturbation growth and compare linearized and nonlinear dynamics, evaluating stability conditions using the Routh criterion and numerical accuracy through Jacobi integral conservation.

PUBLICATIONS AND CONFERENCES

- Kumar, R., Bhaumik, Galrani, R., **Francis, A**., and Dhawan, H. (2023). "Optimization of hydrogen fuel cell economy for next generation power supply." *Materials Today: Proceedings*. doi:10.1016/j.matpr.2023.03.257.
- Chawla, S. S., Bhaumik, **Francis, A**., Bhatia, J., and Chaturvedi, S. K. "Case Study: Boeing 747 Avionics Architecture." *Space Habitat Event Abstract Book*, p. 56.

TECHNICAL SKILLS

Programming Language	MATLAB, Python, C, C++
Mission Analysis	STK, GMAT, HYDRA, Ansys Workbench, Ansys Fluent
CAD & Hardware	AutoCAD, Autodesk Fusion 360, Altium Designer, SolidWorks, CATIA
Space Systems	Command and Data Handling (C&DH), UHF Communications, DA Systems, GNU Radio
Tools & Platforms	Linux, Git, Microsoft Office

INTERNSHIPS

Radar and Communications Inter *June 2023 – August 2023*

Indian Air Force, 13 Base Repair Depot, India

- Assisted in the inspection, testing, and maintenance of airborne radar, communication, and avionics systems, gaining hands-on experience in RF equipment calibration, system verification, and fault diagnosis.
- Investigated aircraft flight recorder recovery systems and proposed a conceptual GPS-enabled underwater locating device to improve aircraft localization during maritime recovery operations.

Mechanical Engineering Intern *May 2023 – Sept 2023*

Indian Institute of Technology Ropar, India

- Contributed to the design and analysis of a hybrid renewable energy system integrating solar photovoltaic, micro-hydro, backup generation, and energy storage technologies to evaluate power reliability and energy demand.
- Performed system-level trade studies, energy modeling, and CAD-based design evaluations while collaborating with a multidisciplinary engineering team.

TEACHING EXPERIENCE

Teaching Assistant – Accelerated Introductory Astronomy I *Aug 2026–Present*

Department of Astrophysical and Planetary Sciences, University of Colorado Boulder

Course Instructor: *Professor David Malaspina*

- Independently led two undergraduate laboratory sections totaling approximately 50 students per week, mentored students in observational astronomy and scientific problem-solving, coordinated nighttime telescope sessions, and supported lecture instruction and examination reviews.

Teaching Assistant – Weather and the Atmosphere Laboratory *Jan 2026–May 2026*

Department of Atmospheric and Oceanic Sciences, University of Colorado Boulder

Course Instructor: *Professor Derek Brown*

- Independently managed two laboratory sections comprising 48 students per week, mentoring students in experimental methods, meteorological data analysis, and scientific interpretation while providing individualized guidance.

ACHIEVEMENTS AND LEADERSHIP

- Top 3 Graduate Capstone Projects in Aerospace Engineering at the Engineering EXPO, CU Boulder, USA – 2026
- Top 5 Mission Proposal for Space Mission Development Concept, University of Colorado Boulder, USA – 2025.
- Top 5 Finalist, Major Project category, Capstone Project Exhibition, UPES, Dehradun – 2024.
- Best Avionics Case Study, Sustainability Fair, UPES, Dehradun – 2023.
- Best Instrumentation Presentation, Aircraft Systems and Instruments, UPES, Dehradun – 2023.